

GUARDIAN · DEPLOYMENT REVIEW

Avalanche Deployment Review.

Verification of the USAT deployment, ownership handover, and messaging configuration on Avalanche C-Chain.

CLIENT

USTDO

PROJECT

USAT · Avalanche

REVIEW FIRM

Guardian

REVIEW WINDOW

September 8-23,
2026

FINAL REPORT

September 27,
2026

§ 00

Contents.

13 pages - overview, confidence ranking, scope, 3 findings, disclaimer, and about Guardian.

§ 01	Overview	p. 03
§ 02	Confidence Ranking	p. 04
§ 03	Scope	p. 05
	Deployment Evidence	p. 06-07
§ 04	Findings	p. 08
	Initial Deployment Review · 2 findings	p. 9-10
	Ownership Transfer Review · 1 finding	p. 11-11
§ 05	Disclaimer	p. 12
§ 06	About Guardian	p. 13

Overview.

Guardian reviewed the USAT token and OFT deployment on Avalanche C-Chain, the completed ownership handover, Avalanche-side peer wiring, and the associated fork-test evidence.

REVIEW FIRM	Guardian	TOKEN REPOSITORY	Everdawn-Labs/usdt0-tether-contracts-hardhat
CLIENT	USTD0	OFT REPOSITORY	Everdawn-Labs/usdt0-oft-contracts
PROJECT	USAT · Avalanche Deployment	DEPLOYMENT PRS	#118 · #235
NETWORK	Avalanche C-Chain · Chain ID 43114	TOKEN REPOSITORY REVISION	68c7365bf344e8b8689effd05e58736a22898aaa
REVIEW WINDOW	September 8-23, 2026	GUARDIAN FORK-TEST REVISION	268a7eb28a511d5f62595f7c1a17d6d6d918c2f0
FINAL REPORT	September 27, 2026		

1.1 Confidence Ranking

5/5

The reviewed deployment artifacts, verified proxy state, completed ownership handover, peer configuration, batch replay, and fork-test evidence provide very high confidence in the assessed Avalanche state. Provider-side DVN destination configuration remains required for three Avalanche routes.

Very High Confidence

1 V. LOW	2 LOW	3 MED	4 HIGH	5 V. HIGH
-------------	----------	----------	-----------	--------------

1.2 Findings at a Glance

SEVERITY	TOTAL	PENDING	IN PROG	DECLINED	ACK	PARTIAL	RESOLVED
Critical	0	0	0	0	0	0	0
High	0	0	0	0	0	0	0
Medium	0	0	0	0	0	0	0
Low	1	1	0	0	0	0	0
Info	2	1	0	0	0	1	0

§ 02

Confidence Ranking.

Guardian assigns a project-level confidence score using the scale below. The ranking combines finding severity, remediation outcome, and the reviewer's assessment of latent risk.

CONFIDENCE RANKING	DEFINITION AND RECOMMENDATION	RISK PROFILE
5 Very High Confidence ASSIGNED	Codebase is mature, clean, and secure. No High or Critical vulnerabilities were found. Follows modern best practices with high test coverage and thoughtful design. Recommendation: Code is highly secure at time of audit. Low risk of latent critical issues.	0 High/Critical findings and few Low/Medium severity findings.
4 High Confidence	Code is clean, well-structured, and adheres to best practices. Only 1 Significant issue was uncovered per week. Design patterns are sound, and test coverage is strong. Recommendation: Suitable for deployment after remediations; consider periodic review with changes.	0-1 High/Critical findings per engagement week and little to no Medium severity issues. Varied Low severity findings.
3 Moderate Confidence	Medium-severity and occasional High-severity issues found. Code is functional, but there are concerning areas such as weak modularity or risky patterns. No critical design flaws were identified, though some patterns could lead to edge-case issues. Recommendation: Address issues thoroughly and consider a targeted follow-up audit depending on code changes.	1-2 High/Critical findings per engagement week.
2 Low Confidence	Code shows frequent emergence of Critical/High vulnerabilities. Audit revealed recurring anti-patterns, weak test coverage, or unclear logic. These characteristics suggest a high likelihood of latent issues. Recommendation: Post-audit development and a second audit cycle are strongly advised.	2-4 High/Critical findings per engagement week, or additional unresolved High/Critical findings uncovered in remediation review.
1 Very Low Confidence	Code has systemic issues. Multiple High/Critical findings (≥ 5/week), poor security posture, and design flaws that introduce compounding risks. Safety cannot be assured. Recommendation: Halt deployment and seek a comprehensive re-audit after substantial refactoring.	≥ 5 High/Critical findings and overall systemic flaws.

§ 03

Deployment Scope.

Avalanche-only deployment scope assembled from the initial rollout and ownership-transfer reviews.

3.1 Deployment changes and test evidence

Token PR #118 · Avalanche token deployment · 68c7365bf344e8b8689effd05e58736a22898aaa

OFT PR #235 · Avalanche OFT deployment

Guardian-authored fork tests · PR #242 · 268a7eb28a511d5f62595f7c1a17d6d6d918c2f0

Wiring batch · wiring_batches/usat-5.json · 9ad18d70e9f7aa3ef1064db923480f2345d6fd82

3.2 Avalanche contracts and authority

Token proxy 0x9A5926fd88F747c357738b1e82aFB46AB80f0A6b

Token implementation 0xD8479f87686ed263D00Ca7505F86327dbeD4171A

OFT proxy 0xDCee5eF6779567C27e69D57f6A143449372D1784

OFT implementation 0xE43e629D4C4e11D12ED98b4A14F9e0290f67436a

Owner and delegate 0x62b3a0f6ffc5efd7692053A9040fE44F9AC8c5Cb

§ 03

Avalanche Deployment State.

Verified contract references and administrative state at the reviewed Avalanche snapshot.

CHAIN ID	AUTHORITY
43114	USAT multisig

3.2.1 Deployed contracts

TEXT	
1	USAT token proxy 0x9A5926fd88F747c357738b1e82aFB46AB80f0A6b
2	USAT token implementation 0xD8479f87686ed263D00Ca7505F86327dbeD4171A
3	USAT OFT proxy 0xDCee5eF6779567C27e69D57f6A143449372D1784
4	USAT OFT implementation 0xE43e629D4C4e11D12ED98b4A14F9e0290f67436a

3.2.2 Ownership and wiring

- The token owner, OFT owner, both ProxyAdmin owners, and Endpoint delegate are the USAT multisig 0x62b3a0f6ffc5efd7692053A9040fE44F9AC8c5Cb .
- The Avalanche token points to the reviewed OFT proxy.
- Avalanche-side peers are configured for the reviewed reciprocal routes.
- The original deploying-EOA state was historical and was superseded by the ownership-transfer review.

§ 03

Fork and Batch Verification.

The supplied evidence separates historical deployment checks, local wiring rehearsal, and observed-state route readiness.

HISTORICAL AVALANCHE CHECKS	BATCH CALLS	ROUTE CONFIGURATIONS	AFFECTED QUOTE PATHS
11/11	12/12	6/6	3

3.3.1 Successful verification

- Historical Avalanche deployment checks passed **11/11**.
- All 12 calls in `usat-5.json` replayed successfully on local forks.
- The replay established reciprocal peers and passed all six route-configuration checks.
- The OFT build, token build, deployment provenance, initializer, token association, and ownership assertions passed.

3.3.2 Recorded quote failures

Messaging quotations remain unavailable for Ethereum → Avalanche, Celo → Avalanche, and Avalanche → Celo because Anchorage DVN destination gas configuration is zero at the reviewed snapshots. The wiring batch configures OApps and Endpoints but cannot configure the provider contracts.

§ 04

Findings.

The source reviews record one Low and two Informational findings in the Avalanche scope. I-01 is Partially Resolved; I-02 and L-01 are Pending.

ID	TITLE	CATEGORY	SEVERITY	STATUS	PAGE
INITIAL DEPLOYMENT REVIEW					
I-01	Avalanche OFT rollout remains incomplete	Warning	INFO	PARTIALLY RESOLVED	p. 9
I-02	Anchorage fees block three USAT routes	Configuration	INFO	PENDING	p. 10
OWNERSHIP TRANSFER REVIEW					
L-01	Missing DVN settings block USAT routes	Configuration	LOW	PENDING	p. 11

Avalanche OFT rollout remains incomplete.

CATEGORY Warning LOCATION deploy/USATDeployments.ts:20-30,46-63 · layerzero-usat.config.ts:6-19,77-115

REVIEW Initial Deployment Review

DESCRIPTION

An operational OFT rollout requires the proxy deployment and token association as well as reciprocal peers, an approved LayerZero security and execution policy and assignment of administrative control. PR #235 deploys the Avalanche `0Upgradeable` proxy with the intended token and Endpoint V2 addresses. However, `layerzero-usat.config.ts` still defines only Ethereum and Celo. Its only generated connection is between those two chains.

The corresponding live configuration is also incomplete. The Avalanche OFT returns a zero peer for Ethereum EID `30101` and Celo EID `30125`. The Ethereum and Celo OFTs likewise return a zero peer for Avalanche EID `30106`. The Avalanche Endpoint currently resolves its default send and receive libraries for both destinations while `enforcedOptions` is empty. The reviewed changes do not specify an approved set of send and receive libraries, ULN and DVN parameters, confirmation counts, executor settings or enforced execution options for either Avalanche route.

Administrative control also remains with the deployment EOA `0x1a6362AD64ccFF5902D46D875B36e8798267d154`. This address is the live OFT owner, ProxyAdmin owner and LayerZero Endpoint delegate. PR #235 does not include the transactions that will transfer these roles or the transactions that will establish and configure the routes. The separate validation changes in PR #237 treat Avalanche messaging configuration as absent, so they check the deployed contracts but skip the messaging behavior.

The deployed creation inputs match the PR artifacts and the token already points to the OFT proxy. Therefore, this issue does not indicate a defect in the deployed bytecode or proxy initialization. It means that the Avalanche OFT cannot send or receive through the intended routes and that the intended final security and control state cannot be reproduced from the reviewed changes.

RECOMMENDATION

Complete the rollout through one reviewed configuration and ownership plan. The plan should establish the reciprocal Avalanche peers on every intended remote chain, apply the approved send and receive libraries, ULN and DVN parameters, confirmations, executor settings and enforced options for each direction then transfer the OFT owner, ProxyAdmin owner and Endpoint delegate to the intended multisig. Record the exact transactions and required final state so the deployment is not considered complete until every route and authority matches that plan.

RESOLUTION

Status: Partially Resolved. At the reviewed Avalanche snapshot, the token owner, OFT owner, both ProxyAdmin owners, and Endpoint delegate are the USAT multisig `0x62b3a0f6ffc5efd7692053A9040fE44F9AC8c5Cb`. Avalanche-side peers are configured, and the wiring batch passes all six route-configuration checks. The remaining Anchorage DVN condition is recorded separately in I-02 and L-01.

INFO I-02

PENDING

Anchorage fees block three USAT routes.

CATEGORY Configuration

LOCATION layerzero-usat.config.ts:59,74-84,100-112 · node_modules/@layerzerolabs/lz-evm-messagelib-v2/contracts/uIn/dvn/DVNFeeLib.sol:129-134

REVIEW Initial Deployment Review

DESCRIPTION

The USAT messaging configuration requires Anchorage alongside LayerZero Labs and USTDO. At the reviewed snapshots, Anchorage's destination configuration has a zero gas value for Ethereum → Avalanche (EID 30106), Celo → Avalanche (EID 30106), and Avalanche → Celo (EID 30125).

When a user requests a messaging quote, the send library queries each required DVN. Anchorage passes its destination configuration to `DVNFeeLib.getFee()`, which reverts with `DVN_EidNotSupported` when `gas == 0`. The same guard is reached when assigning the DVN's send job.

The 12 calls in `wiring_batches/usat-5.json` configure the OFTs and Endpoints without updating Anchorage's destination settings. This provider activation dependency can be part of the staged rollout. Until the provider configuration is completed, ordinary and composed messaging quotes fail for the three directions and the affected sends cannot complete, even after the wiring batch executes successfully.

RECOMMENDATION

Coordinate with Anchorage's authorized DVN administrators to configure the destination gas and fee parameters for EID 30106 on Ethereum and Celo and EID 30125 on Avalanche before opening these routes to users.

RESOLUTION

Status: Pending.

LOW L-01

PENDING

Missing DVN settings block USAT routes.

CATEGORY Configuration REVIEW Ownership Transfer Review

DESCRIPTION

The supplied USAT wiring batch configures Anchorage as one of three required DVNs for the Avalanche connections. However, at the reviewed fork blocks, the corresponding Anchorage DVN contracts have zero destination gas settings for Ethereum → Avalanche, Celo → Avalanche, and Avalanche → Celo.

After successfully replaying the batch, fee quotations for these routes revert with `DVN_EidNotSupported`. The DVN fee library rejects destinations whose configured gas is zero. USAT transfers cannot be initiated through the three affected routes until the required DVN destination configuration is completed.

RECOMMENDATION

Coordinate with the Anchorage DVN administrators to configure the affected destinations through `setDstConfig`.

RESOLUTION

Status: Pending.

§ 05

Disclaimer.

This report is not, nor should be considered, an "endorsement" or "disapproval" of any particular project or team. This report is not, nor should be considered, an indication of the economics or value of any "product" or "asset" created by any team or project that contracts Guardian to perform a security assessment. This report does not provide any warranty or guarantee regarding the absolute bug-free nature of the technology analyzed, nor do they provide any indication of the technologies proprietors, business, business model or legal compliance.

This report should not be used in any way to make decisions around investment or involvement with any particular project. This report in no way provides investment advice, nor should be leveraged as investment advice of any sort. This report represents an extensive assessing process intending to help our customers increase the quality of their code while reducing the high level of risk presented by cryptographic tokens and blockchain technology.

Blockchain technology and cryptographic assets present a high level of ongoing risk. Guardian's position is that each company and individual are responsible for their own due diligence and continuous security. Guardian's goal is to help reduce the attack vectors and the high level of variance associated with utilizing new and consistently changing technologies, and in no way claims any guarantee of security or functionality of the technology we agree to analyze.

The assessment services provided by Guardian is subject to dependencies and under continuing development. You agree that your access and/or use, including but not limited to any services, reports, and materials, will be at your sole risk on an as-is, where-is, and as-available basis.

Cryptographic tokens are emergent technologies and carry with them high levels of technical risk and uncertainty. The assessment reports could include false positives, false negatives, and other unpredictable results. The services may access, and depend upon, multiple layers of third-parties.

Notice that smart contracts deployed on the blockchain are not resistant from internal/external exploit. Notice that active smart contract owner privileges constitute an elevated impact to any smart contract's safety and security. Therefore, Guardian does not guarantee the explicit security of the audited smart contract, regardless of the verdict.

About Guardian.

Founded in 2022 by DeFi experts, Guardian is a leading audit firm in the DeFi smart contract space. With every audit report, Guardian upholds best-in-class security while achieving our mission to relentlessly secure DeFi.